The six-equation two-phase model as implemented in CAMR

Governing equations, closures and source terms

Generated from `CAMR` branch `co2-eos`

Revised for the shared release

Abstract

This note records the two-phase model *as the code actually solves it*, rather than in its textbook form. Each equation, closure and source term is anchored to `file:line`. Runtime options are given with their defaults, because several of the model's decisions are dials rather than constants. Two places where the implementation is known to depart from the model it intends are flagged explicitly in §7.

1 State vector

The conserved state carries the two-phase slots `Utils/IndexDefines.H:145--149` alongside CAMR's ordinary single-fluid slots `:17--18, 74--76`:

$$\mathbf{U} = \big(\, \underbrace{\rho,\ \rho\mathbf{u},\ \rho E,\ \rho e,\ T}_{\text{stock CAMR}},\ \underbrace{\alpha_1,\ \alpha_1\rho_1,\ \alpha_2\rho_2,\ \alpha_1\rho_1 E_1,\ \alpha_2\rho_2 E_2}_{\text{PS extension}} \,\big)^{\top}. \tag{1}$$

Write $m_k = \alpha_k\rho_k$ for the partial masses and $\mathcal{E}_k = \alpha_k\rho_k E_k$ for the phase total energies, with $E_k = e_k + \frac{1}{2}|\mathbf{u}|^2$. The volume fractions close as $\alpha_1 + \alpha_2 = 1$, so only α_1 is stored.

The state is deliberately redundant. Mixture mass and energy are stored *both* as ρ, ρE *and* as the phase sums. The two identities

$$\rho \;=\; m_1 + m_2, \qquad \rho E \;=\; \mathcal{E}_1 + \mathcal{E}_2 \tag{2}$$

are therefore invariants to be *maintained*, not consequences of the discretisation. They are re-established every step by `ps_resync_mixture_mass` and `ps_resync_phase_energy` `CAMR_advance.cpp:410--413` and monitored as checks V4 and V5 of `ps_validate_state`. Historically the mass identity drifted at $\sim$0.4 % per production run before anyone was watching it.

2 Governing equations

With a single mixture velocity $\mathbf{u}$ (velocity equilibrium) and instantaneous mechanical relaxation, the system solved is

$$\frac{\partial \alpha_1}{\partial t} + \mathbf{u}\cdot\nabla\alpha_1 = \mathcal{S}_\alpha, \tag{3}$$

$$\frac{\partial m_1}{\partial t} + \nabla\cdot\big(m_1\mathbf{u}\big) = \Gamma, \tag{4}$$

$$\frac{\partial m_2}{\partial t} + \nabla\cdot\big(m_2\mathbf{u}\big) = -\,\Gamma, \tag{5}$$

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot\big(\rho\,\mathbf{u}\otimes\mathbf{u} + P\,\mathsf{I}\big) = \mathbf{0}, \tag{6}$$

$$\frac{\partial \mathcal{E}_1}{\partial t} + \nabla\cdot\big(\mathcal{E}_1\mathbf{u} + \alpha_1 P_1\mathbf{u}\big) = -\,P_I\,\frac{\partial\alpha_1}{\partial t} \;+\; \mathcal{Q} \;+\; \Gamma H_I, \tag{7}$$

$$\frac{\partial \mathcal{E}_2}{\partial t} + \nabla\cdot\big(\mathcal{E}_2\mathbf{u} + \alpha_2 P_2\mathbf{u}\big) = +\,P_I\,\frac{\partial\alpha_1}{\partial t} \;-\; \mathcal{Q} \;-\; \Gamma H_I, \tag{8}$$

together with the mixture total energy in strict conservation form,

$$\frac{\partial(\rho E)}{\partial t} + \nabla\cdot\big[(\rho E + P)\,\mathbf{u}\big] \;=\; 0, \tag{9}$$

which is the quantity that is fluxed and refluxed, with (2) tying it to (7)–(8). The mixture pressure is the volume-fraction average

$$P \;=\; \alpha_1 P_1 + \alpha_2 P_2, \qquad \texttt{PS\_hllc.H:289} \tag{10}$$

and the interfacial pressure is the arithmetic mean

$$P_I \;=\; \tfrac{1}{2}\,(P_1 + P_2). \qquad \texttt{hem\_pelanti\_shyue.H:2157} \tag{11}$$

Equations (3), (7) and (8) are *not* in divergence form: α_1 is advected rather than fluxed, and the phase energies carry the non-conservative product $P_I\,\partial_t\alpha_1$. That is the entire reason the code uses wave propagation rather than flux differencing for those three slots (§6).

3 Equation of state and the branch-locked contract

The reference backend is Peng–Robinson for pure CO_2 ($T_c = 304.13$ K, $P_c = 7.3773\times10^6$ Pa, $\omega = 0.22394$). Queries are *branch-locked*: the caller nominates the phase, and the EOS returns that branch, continued metastably past its saturation limit if necessary. It never auto-detects, because a nominal vapour sitting at compressed-liquid density would otherwise return the liquid root and drive the relaxation toward an unphysical equilibrium.

The corresponding anti-requirement is load-bearing:

> *the hydraulic path must not ask the EOS about a mixture state.*

For a two-phase cell the pair (ρ, e) combines the light phase's volume with the heavy phase's mass and need not be a single-fluid state at all, so the inversion may have no root. Temperature inversion brackets $T \in [1, 5000]$ K and *refuses* outside it rather than returning the nearest reachable state.

4 Source terms

The default relaxation is the **coupled X3 operator** (`ps_relax_mode`$=5$, the default and the acceptance configuration since 2026-08-17; an unrecognised mode *aborts* rather than falling through `PS_relaxation.H:1030--1050`). Per reaction substep `CAMR_advance.cpp:640--665`:

$$\text{hydro} \longrightarrow \underbrace{\text{vanish fold}}_{\text{phase death}} \longrightarrow \underbrace{\text{X3}}_{\substack{P_1=P_2\ \text{DAE} + \text{thermal} +\\ \text{BE-SRT mass transfer}}} \longrightarrow \underbrace{\text{flash}}_{\text{nucleation}} \longrightarrow \underbrace{\text{mech. reproject}}_{\texttt{ps\_src\_p\_reproject=1}} \tag{12}$$

X3 solves the cell as a constrained system: mechanical equilibrium $P_1 = P_2$ is imposed as a DAE constraint while the thermal and mass-transfer legs run as backward-Euler rates *on that constrained manifold* — one coupled solve, not a sequence, so the operators cannot fight over α_1. Its 0-D fixed point is the full-equilibrium (HEM) flash. A closing mechanical pass (`ps_mech_close`= 1 default) re-establishes $P_1 = P_2$ at chain exit. The historical sequential chain (mechanical → thermal → flash → MT with a re-projection after each of the last two) survives as `ps_relax_mode`= 4; mode 0 is mechanical-only and must now be asked for explicitly. In mode 5 the split mass-transfer source is *owned by X3*: enabling it separately aborts `CAMR_advance.cpp:139--147`.

4.1 Mechanical relaxation: $\mathcal{S}_\alpha$

Pressure equilibrium is **instantaneous by model closure**: α_1 is the free variable and is driven until $P_1 = P_2$. It is a constraint, not a rate. The accompanying interfacial work is applied to the phase energies `hem_pelanti_shyue.H:2223--2225`:

$$e_1 \ \mapsto\ e_1 - \frac{P_I\,\Delta\alpha_1}{\alpha_1\rho_1}, \qquad e_2 \ \mapsto\ e_2 + \frac{P_I\,\Delta\alpha_1}{\alpha_2\rho_2}, \tag{13}$$

equal and opposite in $\mathcal{E}_k$, so mixture energy is preserved exactly. This is the discrete realisation of $\mp P_I\partial_t\alpha_1$ in (7)–(8). The Newton step on α_1 is capped at a fraction of $\min(\alpha_1, 1-\alpha_1)$ and restricted to the interval where both ρ_k remain inside the EOS domain.

4.2 Thermal relaxation: $\mathcal{Q}$

Finite-rate, isochoric, at the α_1 the mechanical pass chose. It redistributes the energy split only, driving $T_1 \to T_2$ at rate $1/\theta$ `hem_pelanti_shyue.H, ps_iso_thermal_relax_cell_finite`:

$$\mathcal{E}_k^{n+1} = \mathcal{E}_k^{*} + \left(\mathcal{E}_k^{\mathrm{eq}} - \mathcal{E}_k^{*}\right)\left(1 - e^{-\Delta t/\theta}\right), \tag{14}$$

where $\mathcal{E}^{\mathrm{eq}}$ is the isochoric $T_1 = T_2$ state. Note the form: this is an *exponential relaxation onto a precomputed equilibrium.* **`CAMR.ps_theta_tau` defaults to 10^{-7} s** (THETA-DEFAULTS 2026-08-24) — near-instantaneous at hydrodynamic scales — and its semantics are: $\theta > 0$ a finite backward-Euler rate; $\theta \le 0$ *instantaneous* thermal equilibrium, i.e. in mode 5 the constraint $T_1 = T_2$ joins the $P_1 = P_2$ DAE ("$\theta \le 0$ does not mean off" `PS_relaxation.H:1139--1151`).

The leg is gated by a two-phase coexistence test — both phase temperatures strictly inside $(T_{\mathrm{triple}}, T_{\mathrm{crit}}) = (216.592, 304.13)$ K — shared with mass transfer and the flash. `CAMR.ps_coexist_action` (default 0) selects what happens when a phase leaves that band: 0 silent no-op, 1 abort, 2 run the thermal leg for low-side exits, 3 run it for any exit. The gate carries a **runaway clause** `hem_pelanti_shyue.H, PS_relaxation.H`: a *high-side-only* band exit with a phase beyond $2\,T_{\mathrm{crit}}$ (608 K for CO_2; the factor is a constexpr on the EOS-supplied T_{crit}, so it is fluid-agnostic) runs the thermal leg regardless, because the alternative — vetoing thermal while the mechanical projection keeps pumping interfacial work into a vanishing sliver — was measured to ratchet e_2 to the EOS band edge at the jet contact. Temperature is the discriminator because it separates the two populations with wide margins: the hottest legitimate phase in the validated suite is B10's vapour at 400 K $= 1.32\,T_{\mathrm{crit}}$, whereas the runaway ratchets through $2\,T_{\mathrm{crit}}$ on its way to $16\,T_{\mathrm{crit}}$; a mass-trace discriminator was measured and rejected, since B4/B10's own smeared contact edges hold trace-mass cells below the sliver's mass ratio. Material cross-critical contacts ($\lesssim 1.5\,T_{\mathrm{crit}}$) keep their veto. The clause is bit-inert on the 1-D battery and drains the sliver periodically on the flashing-vent reference instead of the monotone ratchet. Runaway activations are counted (`n_exit_runaway`, exit-code bit 8) and printed in `[PS-COEXIT-TH]`.

(The per-cell HRM-correlation option `CAMR.ps_theta_model`= 1 — Downar-Zapolski $\theta = \theta_0\alpha_v^{-0.54}\psi^{-1.76}$ — was **retired** 2026-08-17 (S3): measured to be the wrong mapping for *thermal* contact between phases, since the correlation slows toward saturation, which is right for phase change and wrong for heat exchange; B9 aborts under it. The correlation itself remains relevant to mass transfer; see §7(ii).)

4.3 Mass transfer: Γ and H_I

Driven by the Gibbs free-energy difference $g_1 - g_2$ with $g_k = h_k - T_k s_k$, and integrated as an exact relaxation onto the instantaneous mass-transfer equilibrium:

$$\delta m \;=\; \left(1 - e^{-\Delta t/\tau}\right)\delta m_{\mathrm{eq}}, \qquad \delta m_{\mathrm{eq}}:\; g_1 = g_2, \tag{15}$$

so $\Gamma = -\delta m/\Delta t$ in (4). The target δm_{eq} is found by a Newton iteration on δm. The form (15) is unconditionally stable and cannot overshoot δm_{eq} — *provided* the target is the equilibrium of the path actually integrated; see §7.

The default rate is derived, not tuned (BE-SRT). At the default (mode 5, `ps_mt_tau_model`= 2, ASY1) no relaxation time is chosen: τ in (15) is *computed* as the distance to equilibrium over the instantaneous physical rate, per Lund & Aursand,

$$\tau_{\mathrm{asy}} = \frac{|\delta m_{\mathrm{eq}}|}{\Gamma_{\mathrm{SRT}}}, \qquad \Gamma_{\mathrm{SRT}} = \Sigma\,\rho_g\sqrt{\frac{M}{2\pi R_u T_g}}\,\big|g_1 - g_2\big|, \qquad \Sigma = \frac{16}{\pi}\,\frac{(\alpha_{\mathrm{recv}} + \delta)\,\alpha_{\mathrm{send}}}{D}, \tag{16}$$

`hem_pelanti_shyue.H:2364--2407`: the kinetic prefactor is parameter-free statistical rate theory; the *interfacial-area* closure Σ is not — it carries the stratified-pipe morphology placeholder D = `CAMR.ps_mt_srt_d` (default 0.1 m) and the receiving-phase seed δ = `CAMR.ps_mt_srt_delta` (default 0.01). A degenerate Γ falls back to the constant-τ form so the dial can never invent a rate. The morphology dependence this hides is quantified in §7(ii). The constant `ps_mt_tau` route belongs to the non-default modes.

Transferred mass carries the interface total enthalpy

$$H_I = h_I + \tfrac{1}{2}|\mathbf{u}|^2, \qquad h_I = \begin{cases} (1-w)\,h_1 + w\,h_2, & w \ge 0 \quad (w = \tfrac{1}{2} \text{ default: mean}),\\ h_1 \text{ if } \delta m > 0, \quad h_2 \text{ otherwise}, & w < 0 \quad (\text{upwind}), \end{cases} \tag{17}$$

selected by `CAMR.ps_mt_h_weight`.

Volume co-moves with the mass (E.1). Under the presence model the transfer moves volume at the *donor's own* density:

$$\Delta\alpha_1 \;=\; -\frac{\delta m}{\rho_1}. \tag{18}$$

Two consequences make this the derived choice rather than a convenience. The donor's density is then *exactly* invariant, $\rho_1' = (m_1 - \delta m)/(\alpha_1 - \delta m/\rho_1) = \rho_1$, so it can never leave the EOS domain from below; and the receiver's density becomes a convex combination (a mediant) of ρ_2 and ρ_1, so it can never leave from above. The proof is rate-independent: even $\delta m = m_1$ in one step lands α_1 exactly on zero.

4.4 Flash: nucleation only

The flash is **not** a conversion operator. It seeds a minority phase up to $\alpha_{\text{birth}} = 4 \times 10^{-2}$ (`PsPres::alpha_birth`, single-sourced from the presence parameters, §5) and declines once that level is reached `hem_pelanti_shyue.H, ps_flash_source_cell`:

$$\alpha_2 \;\mapsto\; \alpha_2 + \beta\left(\alpha_{\text{birth}} - \alpha_2\right), \qquad \text{return if } \alpha_2^{\text{new}} \le \alpha_2, \tag{19}$$

with β a blend of a metastability window, a dominance ramp and $\Delta t/\tau_f$. Newborn mass is injected at the *saturation* density ρ_{sat}, which is (18) run backwards, so birth and death are the same operator with opposite sign. Past α_{birth}, conversion is mass transfer's job alone.

5 Presence: when a phase has a state at all

The intensive quantities are quotients with *different* denominators,

$$\rho_k = \frac{m_k}{\alpha_k}, \qquad e_k = \frac{\mathcal{E}_k}{m_k} - \tfrac{1}{2}|\mathbf{u}|^2, \tag{20}$$

so a test on one does not license the other. Because the hydro advances m_k with an absolute error set by the *mixture* scale, ρ_k inherits a relative error $\sim \eta/\alpha_k$. A phase is therefore assigned one of three regimes by α_k alone — no auxiliary flag array, so the regime cannot fall out of step with the state:

regime	α_k	meaning
ABSENT	exactly 0	no intensive quantity is ever formed; the cell is single-phase Euler
CORRIDOR	$(\alpha_{\text{van}}, \alpha_{\text{cond}})$	transport-only: advects, but no EOS query, no relaxation, no transfer
INDEPENDENT	$[\alpha_{\text{cond}}, 1-\alpha_{\text{cond}}]$	a full six-equation phase; every operator acts

with $\alpha_{\text{cond}} = 2 \times 10^{-2}$ (a *conditioning* bound, re-derived 2026-08-15 as $\eta_{\max}/\varepsilon$ from a measured per-step $\eta_{\max} = 5.2 \times 10^{-4}$ at a 5 % accuracy target), $\alpha_{\text{van}} = 10^{-8}$ (death), and $\alpha_{\text{birth}} = 2\alpha_{\text{cond}}$ (nucleation seed). The ordering gives hysteresis with no flip-flop channel: birth at $2\alpha_{\text{cond}}$, loss of independence only below α_{cond}, death only below α_{van}.

Promotion degeneracy floor and relax-gate hysteresis. Volume fraction alone does not license promotion. In the corridor α_1 is advected non-conservatively at S_M while m_1 is fluxed conservatively and no source re-couples them, so the two slots drift apart freely in rough flow; a phase can arrive at $\alpha_k \ge \alpha_{\text{cond}}$ with $\rho_k = m_k/\alpha_k$ far below anything its branch can answer (measured: a promotion at $\rho_1 = 0.076$ kg/m^3, $e_1 = 7.6 \times 10^7$ J/kg, which the branch-locked inversion rightly refused). The regime classifier `ps_regime` therefore requires, in addition, $m_k > \rho_{\text{deg}}\,\alpha_k$ (division-free) with ρ_{deg} = `CAMR.ps_presence_rho_deg` = 0.5 kg/m^3; otherwise the phase stays CORRIDOR whatever its α_k. The value is fixed by two measurements: the minimum legitimate independent-phase density over all battery finals is 1.406 kg/m^3 (C3) and the degenerate promotion carried 0.076, so 0.5 separates them by a factor three each way, and zero battery-final cells trip it. The same floor feeds two conservative reaps in the vanish fold (a corridor phase with $m_k \le \rho_{\text{deg}}\alpha_k$, or a phase with e_k outside [`ps_presence_e_deg_lo`, `ps_presence_e_deg_hi`] = $[-2 \times 10^6, 2 \times 10^7]$ J/kg, is folded into its host with mass and energy conserved; counted `n_cdeg`/`n_edeg` in `[PS-FOLD]`). Checked promotion (`CAMR.ps_promote_checked`= 1, `PS_promote.H`) additionally demotes a would-be independent phase whose energy quotient defines no state on its own branch. Finally the relaxation gate carries hysteresis (`CAMR.ps_relax_hyst`= 1, `ps_presence_relax_gate`): a phase in the upper corridor, $\alpha_k \ge \alpha_{\text{cond}}/2$ with $m_k > \rho_{\text{deg}}\alpha_k$, still counts as relaxable, so the source on/off boundary no longer coincides with the exact α the relaxation itself moves (the driver of near-threshold regime toggling in 2-D production). The floor and reaps are bit-inert on the 1-D battery; the hysteresis moves four near-threshold cases by at most +6.7 % in a P-error norm (B11) and is kept for the 2-D behaviour; `ps_relax_hyst`= 0 is the one-key A/B. The hysteresis band is evaluated on the host only.

6 Discretisation of the non-conservative slots

Since 2026-08-27 the interior scheme is **single-path**: wave propagation (`CAMR.ps_flux=wp`, the Berger–LeVeque fluctuation form) is the only interior flux. `llf`, `hllc`, a set `ps_ctu`, `ps_recon` or `ps_alpha_limiter` key all *abort* with retirement notes `PS_umeth.cpp:5--12,727--745`; `PS_reconstruction.H` is deleted — the wp interior works from raw cell averages. Under AMR the fluctuation form refluxes through the standard register with α moved alongside (`CAMR.ps_bl_reflux`= 2, the default since 2026-08-26).

One Riemann solve per face produces a wave decomposition (retained in `PS_hllc.H`, which now serves the wp fluctuations), from which the code takes *two different exits.*

Conserved slots $(\rho, \rho\mathbf{u}, \rho E, m_1, m_2)$ receive a single recovered interface flux $F^\star = F(\mathbf{U}_L) + \mathsf{A}^-\Delta Q$, which telescopes exactly to the fluctuation update, so CAMR's ordinary flux-difference path and its AMReX flux register are used unchanged.

Non-conserved slots $(\alpha_1, \mathcal{E}_1, \mathcal{E}_2)$ get a face flux of *exactly zero* and a direct fluctuation deposit,

$$\left.\frac{\mathrm{d}\alpha_1}{\mathrm{d}t}\right|_i = -\frac{\mathsf{A}^+\Delta Q_{i-1/2} + \mathsf{A}^-\Delta Q_{i+1/2}}{\Delta x}, \tag{21}$$

and likewise for $\mathcal{E}_1, \mathcal{E}_2$. Because both exits are fed by the same wave structure they cannot disagree about what happened at the face, which is what makes a uniform-pressure, uniform-velocity interface drift without generating spurious waves *by construction* rather than by cancellation between two independent discretisations.

6.1 Wave structure

Three waves per face (HLLC, specialised to the six-equation system following Pelanti 2022). Outer speeds are Davis estimates using each side's own mixture sound speed `PS_hllc.H:375--376`:

$$S_L = \min\big(u_L - c_L,\ u_R - c_R\big), \qquad S_R = \max\big(u_L + c_L,\ u_R + c_R\big), \tag{22}$$

with the Wallis frozen mixture speed `PS_hllc.H:303`

$$c^2 = Y_1 c_1^2 + Y_2 c_2^2, \qquad Y_1 = \frac{\alpha_1\rho_1}{\rho}, \quad Y_2 = 1 - Y_1. \tag{23}$$

"Frozen" means no mass transfer during the acoustic response; the much slower equilibrium speed belongs to the relaxation operator, which is applied separately. (`CAMR.ps_cmix_model`, which offered $c = \max(c_1, c_2)$ and the Wood equilibrium form, was measured 2026-08-13 and **retired** 2026-08-17: MAX gains 3 % on a contact because S_L, S_R only bound the fan, and WOOD aborts B9 with a real-fluid EOS `PS_presence.H:127--136`. The frozen mass-weighted form is now the only form.)

The contact speed follows from the mixture mass and momentum balance `PS_hllc.H:386`:

$$S_M = \frac{P_R - P_L + \rho_L u_L(S_L - u_L) - \rho_R u_R(S_R - u_R)}{\rho_L(S_L - u_L) - \rho_R(S_R - u_R)}. \tag{24}$$

α_1 is advected at S_M — the local contact velocity at that face, not a cell-averaged mixture velocity — which is what preserves the contact.

6.2 Star states

Across each acoustic wave the mixture Rankine–Hugoniot conditions fix one contraction ratio per side `PS_hllc.H, ps_star_state`:

$$r_K = \frac{S_K - u_K}{S_K - S_M}, \qquad m_1^\star = r_K\, m_1, \qquad m_2^\star = r_K\, m_2, \qquad u^\star = S_M. \tag{25}$$

The partial-mass scaling is *forced*: with one velocity and α jumping only at the contact, per-phase mass conservation across a wave moving at S_K gives $m_k^\star(S_K - S_M) = m_k(S_K - u_K)$ for each phase separately, so composition is untouched by the acoustic compression. What the mixture algebra does *not* fix is the volume partition. The frozen closure (Pelanti 2022, eq. B.14) freezes α across acoustic waves, which forces *equal volumetric strain* $\rho_k^\star = \rho_k r_K$ on both phases; the model's own closure then repairs that intermediate by instantaneous mechanical relaxation immediately afterwards. A corridor phase (§5) is handed the compression but denied the repair, and the equal strain over-compresses the stiff phase by an order of magnitude at a two-phase wall reflection. The code therefore embeds the relaxed intermediate in the star state itself.

The relaxed-α construction. Behind each wave the two phases sit at mechanical equilibrium at a common pressure rise $\mathrm{d}P$, and each takes the strain its own bulk modulus $B_k = \rho_k c_k^2$ admits:

$$s_k = \frac{\mathrm{d}P}{B_k}, \qquad \rho_k^\star = \rho_k\,(1 + s_k), \qquad \alpha_k^\star = \frac{m_k^\star}{\rho_k^\star} = \frac{\alpha_k\, r_K}{1 + s_k}, \tag{26}$$

and volume saturation $\alpha_1^\star + \alpha_2^\star = 1$ becomes one scalar equation in $\mathrm{d}P$,

$$f(\mathrm{d}P) = \frac{\alpha_1 r_K}{1 + \mathrm{d}P/B_1} + \frac{\alpha_2 r_K}{1 + \mathrm{d}P/B_2} - 1 = 0, \tag{27}$$

which is monotone decreasing wherever $1 + s_k > 0$, so its root is unique. The seed is the linearisation of (27), $\mathrm{d}P_0 = (1 - 1/r_K)\big/\big(\alpha_1/B_1 + \alpha_2/B_2\big)$ — the mixture volumetric strain distributed by the Wood (mechanical-equilibrium) compressibility, which is precisely what "relaxed" means — followed by *three* Newton steps, a fixed count with no convergence threshold to tune. The energy partition keeps the kinetic term and pays each phase $P\,\mathrm{d}v$ for its *own* volume change,

$$E_k^\star = E_k + (S_M - u_K)\, S_M + P\, \frac{s_k}{\rho_k\,(1 + s_k)}, \qquad \frac{s_k}{\rho_k(1 + s_k)} = \frac{1}{\rho_k} - \frac{1}{\rho_k^\star}, \tag{28}$$

with P the *mixture* pressure (10). (The per-phase alternative `CAMR.ps_pk_energy_flux= 1` is retired; a set key aborts.) Whenever (27) is solved exactly, $\sum_k m_k^\star E_k^\star = r_K\big[\rho E + \rho(S_M - u_K)S_M\big] + P\,(r_K - 1)$, term for term the B.14 mixture sum, so the mixture energy Rankine–Hugoniot condition holds *exactly* and every conserved-slot star component (ρ, momenta, ρE, ρe, both partial masses, species) is identical between the two constructions: the conservative flux route is untouched, wave speeds and Δt are untouched, and the change is confined to the three non-conserved slots' wave components, where α_1 now jumps across the acoustic waves too (the Kapila compression term, discretely) and the phase energies are re-partitioned. No new constant enters: c_k, α_k, ρ_k, P, r_K, S_M are all already in the face state.

Limits. Equal bulk moduli ($B_1 = B_2$, which includes every single-phase cell and the c-slaved corridor face with $\rho_1 = \rho_2$) give $1 + s_k = r_K$ for both phases, $\alpha_k^\star = \alpha_k$, $\rho_k^\star = \rho_k r_K$, and the work term reduces to $(S_M - u_K)P/q_k$ with $q_k = \rho_k(S_K - u_K)$: B.14 is the equal-compressibility special case. A zero-strength wave ($S_M = u_K$, $r_K = 1$) gives $\mathrm{d}P = 0$ and star = cell state, so the uniform interface and the identity case are fixed points. As $\alpha_1 \to 0$ the trace phase's strain scales by B_2/B_1; with the host-slaved corridor sound speed that ratio is ρ_2/ρ_1. Exact $\alpha_1 \in \{0, 1\}$ skips the solve, since the forms coincide there and the presence design needs those values preserved bit-exactly. The fan geometry (S_L, S_R from the Wallis frozen speed) stays frozen-acoustic while the internal partition is relaxed; this mirrors what the cell update does (frozen hydro, then instantaneous relaxation) and leaves upwinding and Δt alone.

Degeneracy. A non-finite or vanishing bulk modulus, $1 + s_k \le 0$ at any Newton iterate (a rarefaction past a phase's linear-acoustic range), a non-finite iterate, or a root with $\alpha_1^\star \notin (0,1)$ makes the face fall back to the equal-strain B.14 partition for that face — the model's own weak form, a principled fallback rather than a clamp — and is counted (`face_diag::n_relax_fb`); the measured fallback rate is zero across the 1-D suite and 225 steps of 2-D production. Measured on acceptance: the single-phase A/C battery and the identity case move by exactly nothing; the two-phase wall reflection B12 drops from a reflection ratio $c_1/c_2 = 1.0000$ to 0.0097 (gate ≤ 0.25); the both-phases-independent B5 keeps the energy identity to 6×10^{-14} with zero fallbacks.

Failure is refusal, not repair. A non-finite intermediate, a vanishing denominator or a non-positive star density makes the face return `false`, and that face alone falls back to a local Lax–Friedrichs flux. Nothing is clamped to make the star state exist.

6.3 Second-order correction and the contact-wave taper

At `CAMR.ps_wp_order`= 2 (the acceptance order, set by every deck; the compiled default is 1) each wave carries a van Leer-limited Lax–Wendroff correction flux (`CAMR.ps_wp_limiter`, `vanleer`; `none` gives the unlimited correction for smooth order-of-accuracy tests only). The limiter ratio is one scalar per wave, formed from a nondimensionalised projection of the wave onto its upwind neighbour (`CAMR.ps_wp_proj_scale`= 1), so the per-wave linear identities $W[\rho] = W[m_1] + W[m_2]$ and $W[\rho E] = W[\mathcal{E}_1] + W[\mathcal{E}_2]$ survive limiting: scaling a wave by a scalar preserves them, whereas limiting slots individually would not. The contact wave (S_M) carries the non-conservative α jump, and correcting it at a genuine material interface smears α into the phase densities and energies; dropping the correction everywhere costs single-phase contact sharpness (A/C battery mean 0.0350 → 0.0374). The code therefore scales the whole contact-wave correction by one face weight `PS_umeth.cpp`,

$$w_c = 1 - \mathrm{smoothstep}\left(\frac{|\alpha_{1,L} - \alpha_{1,R}|}{\alpha_{\mathrm{cond}}}\right), \qquad \mathrm{smoothstep}(t) = t^2(3 - 2t), \quad t \in [0,1], \tag{29}$$

(`CAMR.ps_lw_skip_contact`= 2, the default): $|\Delta\alpha| = 0$ — uniform trace or single-phase — keeps the full correction and the A/C mean 0.0350 exactly; $|\Delta\alpha| \ge \alpha_{\mathrm{cond}}$, a material interface, drops it; the smoothstep between removes the on/off kink that a hard regime switch imprints on the front at the α_{cond} contour. The blend width *is* α_{cond}, so no new threshold enters; a level-based taper was rejected because the A/C uniform trace sits below α_{cond} and would be wrongly skipped, while the jump-based taper keeps uniform regions at weight one at any level. On the 2-D jet the taper removes the α_{cond} kink, reduces the near-orifice zigzag about fivefold and halves the spurious extrema. Modes 0 (full correction) and 1 (blanket drop) remain as A/B opt-outs.

7 Two known departures of the implementation from the model

Recorded here because they are properties of the code, not of the model. Both were identified on 13 August 2026; their status as of 28 August:

(i) The mass-transfer target and the step's path (resolved by default). The step applies (18) with the carrier (17); historically δm_{eq} in (15) was computed at *fixed* α_1 with a *mean* carrier regardless of w. Along fixed α_1, $\partial\rho_1/\partial m = 1/\alpha_1$, whereas along (18) it is exactly zero, so the target's Gibbs sensitivity carried a $1/\alpha_1$ amplification the step does not have, breaking the non-overshoot property of (15). `CAMR.ps_mt_target` **now defaults to** 1 (consistent target: same carrier and the same (E.1) path as the step `PS_sources.H:355--362`); 0 retains the legacy behaviour for A/B.

(ii) The mass-transfer rate carries a morphology assumption — now with a measured handle. The physical rate is set by the interfacial area the two phases share *within* a cell; the six-equation state carries no interfacial-area variable, so the SRT closure (16) parameterises it through the single length D. A dispersed mixture and a numerically smeared contact can carry identical cell states and still require rates differing by orders of magnitude; the coexistence gate of §4 acts as a crude binary proxy for that distinction. Two measurements now calibrate the dial (WORKLOG 2026-08-27/28, the DT7-4.1 shock tube of SINTEF report DT7 2019-06): (a) the *effective* HRM relaxation time of the scheme maps linearly onto the report's θ-axis as $\theta_{\mathrm{eff}} \approx 0.08\, D$ [s per m], verified over four decades including the slow-branch topology change, so the default $D = 0.1$ m behaves as $\theta_{\mathrm{eff}} \approx 8$ ms; and (b) the Downar-Zapolski correlation evaluated at that flow's own fan states prescribes $\theta \sim 2 \times 10^{-5}$ s — the default morphology placeholder is ~ 2.5 decades *slower* than the flashing-flow correlation on that problem. A DZ-defensible default would be $D \sim 2.5 \times 10^{-4}$ m; changing it is an open (G-DEF-gated) decision, not yet taken.

8 Characteristic outflow (NSCBC) with a flashing ghost

Subsonic characteristic outflow (`CAMR.ps_bc_use_nscbc`= 1; construction in `PS_nscbc.H`) builds the ghost state from the outgoing invariant R^+ (minmod-extrapolated) and an ambient-encoding target $R^-_{\mathrm{targ}} = -P_{\mathrm{amb}}/\rho c$, with a restoring pull $\sigma = 0.25$; the ambient pressure is `prob.p_amb`, overridable *per face* via `CAMR.ps_bc_p_amb_{x,y,z}{lo,hi}` (sentinel -1 inherits). The per-phase repack integrates the isentrope adaptively in pressure sub-steps capped at $\Delta P \leq 0.05\, \rho_k c_k^2$ (one step reproduces the historical small-signal arithmetic bit-for-bit). For large exterior jumps the ghost is a **choked fan**: the state marches toward P_{amb} along the per-phase isentropes with HEM saturated re-closure inside the dome (T_{sat} by bisection on the EOS P_{sat}; enthalpy-lever quality), $\mathrm{d}u = -\mathrm{d}P/(\rho c_{\mathrm{path}})$ with $c^2_{\mathrm{path}} = \mathrm{d}P/\mathrm{d}\rho_{\mathrm{mix}}$ along the path, terminating at the critical-flow throat ($G = \rho u$ maximum) or at P_{amb}, whichever comes first. `CAMR.ps_bc_nscbc_flash`= 1 (default) enables the in-dome re-closure; 0 reproduces the frozen construction for A/B; other values abort. Measured on the flashing-vent benchmark: vented-mass deficit $-4.1\,\%$ against a boundary-free reference (the retired legacy construction sat at $-7.6\,\%$, the frozen one at $-34\,\%$). The legacy treatment and its `ps_bc_nscbc_v2` key are retired (a set key aborts).

Summary of runtime options referenced above

parameter	default	selects
`ps_flux`	`wp`	the interior scheme (only value; others abort)
`ps_wp_order`	1	2 = limited second-order correction, §6.3 (the acceptance order, set
`ps_lw_skip_contact`	2	contact-wave correction: 0 full, 1 blanket drop, 2 the taper (29)
`ps_bl_reflux`	2	AMR reflux handling of the wp fluctuations (α co-move at the coar
`ps_relax_mode`	5	X3 coupled relaxation (12); 4 = legacy chain
`ps_theta_tau`	10^{-7} s	θ in (14); ≤ 0 = joint T constraint
`ps_mt_srt_d`	0.1 m	morphology length D in (16)
`ps_mt_srt_delta`	0.01	receiving-phase seed δ in (16)
`ps_mt_h_weight`	0.5	the carrier w in (17)
`ps_mt_update_alpha`	on	the (E.1) rule (18)
`ps_mt_target`	1	consistent target, §7(i); 0 = legacy
`ps_coexist_action`	0	response to a band exit (runaway clause always active)
`ps_mech_kernel`	0	mode-0 mechanical kernel; 1 = COTT impedance-weighted
`ps_mech_close`	1	closing mechanical pass at chain exit
`ps_src_p_reproject`	1	re-project $P_1 = P_2$ after flash/MT
`ps_flash_tau`	10^{-7} s	flash nucleator time scale (≤ 0 = off)
`ps_alpha_cond`	2×10^{-2}	the conditioning bound
`ps_alpha_birth`	4×10^{-2}	the flash seed level ($= 2\alpha_{\text{cond}}$)
`ps_presence_vanish`	10^{-8}	the death edge (vanish fold threshold)
`ps_presence_rho_deg`	0.5 kg/m^3	promotion degeneracy floor, §5
`ps_promote_checked`	1	checked promotion to INDEPENDENT (0 = A/B opt-out)
`ps_floor_indep`	1	per-phase floor leg acts on INDEPENDENT phases only (0 = A/B op
`ps_relax_hyst`	1	relax-gate hysteresis band, §5
`ps_bc_use_nscbc`	0	characteristic outflow on every `Inflow`-typed face, §8
`ps_bc_nscbc_{x,y,z}{lo,hi}`	-1	per-face NSCBC enable (0/1; -1 inherits the global key)
`ps_bc_p_amb_{x,y,z}{lo,hi}`	-1	per-face ambient pressure [Pa] (-1 inherits `prob.p_amb`)
`ps_bc_nscbc_flash`	1	HEM re-closure along the boundary fan

Retired keys (a set key aborts with a retirement note). `ps_recon`, `ps_ctu`, `ps_alpha_limiter`, `ps_flux=llf/hllc`, `ps_pk_energy_flux`, `ps_cmix_model`, `ps_theta_model`, `ps_hrm_theta`, `ps_alpha_vanish` (superseded by `ps_presence_vanish`), `ps_bc_nscbc_v2`, `ps_star_relaxed` (the relaxed-α star state of §6.2 is the only construction; = 0 aborts), `ps_rk_model` (a per-phase mass partition that violated the forced per-phase mass jump (25)). Retirement-by-abort rather than silent acceptance is deliberate: a stale deck fails loudly instead of running a configuration that no longer exists.